Experiment No. : 3

Experiment Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Experiment Program Code

Code 1: SampleIF.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h2> If Statement Example</h2>
    <script>
        let age = 20;
        if(age >= 18){
            document.write("You are eligible to vote.")
        }
        else{
            document.write("Not eligible.")
        }
    </script>
</body>
</html>
```

Code 2: While.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h2>While Loop Example</h2>
    <script>
        let i = 1;
        while(i <= 10){
            document.write(i + "<br>");
            i++;
        }
    </script>
</body>
</html>
```

Code 3: MouseUp_Down.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Mouse Down and Mouse Up</title>
</head>
<body>
    <h2>Mouse Down and Mouse Up Events.</h2>
    <div id = "box" style = "width:200px; height:100px; background:skyblue; text-align:centre; line-
height:100px;">Press Mouse</div>

    <script>
        const box = document.getElementById("box");
        box.addEventListener("mousedown", function() {
            box.innerHTML = "Mouse Button Pressed";

        });

        box.addEventListener("mouseup", function(){
            box.innerHTML = "Mouse Button Released";
        });

    </script>
</body>
</html>
```

Code 3: Two_Boxes.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <div id = "box1" style = "width:200px; height:100px; background:Red; text-align:centre;line-
height:100px;">Move Mouse Here</div>

    <div id = "box2" style = "width:200px; height:100px; background:Yellow; text-align:centre;line-
height:100px;">Move Mouse Here</div>

    <script>
        let b1 = document.getElementById("box1");

        let b2 = document.getElementById("box2");

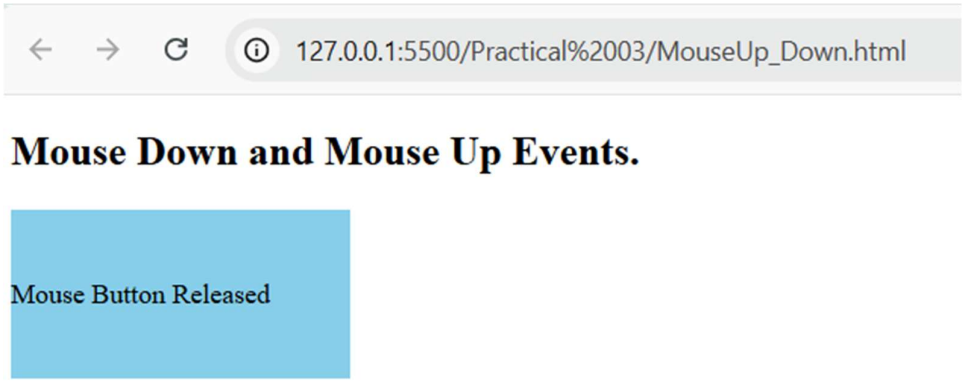
        box1.addEventListener("mouseover", function () {
            box1.style.backgroundColor = "yellow";
            box2.style.backgroundColor = "red";
            box1.innerHTML = "Mouse is Over Box 1!";
            box2.innerHTML = "Move Mouse Here";
        });

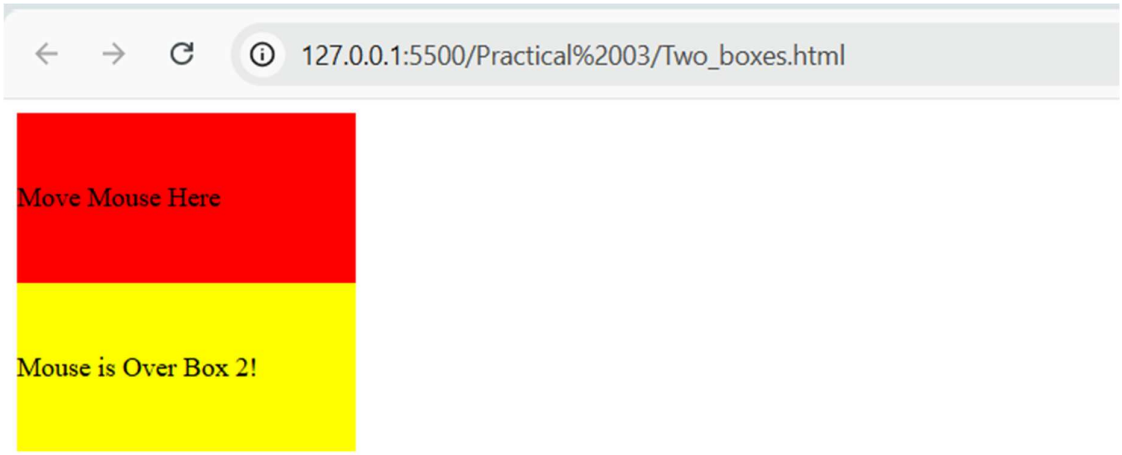
        box2.addEventListener("mouseover", function () {
            box2.style.backgroundColor = "yellow";
            box1.style.backgroundColor = "red";
            box2.innerHTML = "Mouse is Over Box 2!";
            box1.innerHTML = "Move Mouse Here";
        });
    </script>
</body>
</html>
```

4. Output


127.0.0.1:5500/Practical%2003/Do_while.html
1
2
3
4
5
6
7
8
9
10


127.0.0.1:5500/Practical%2003/SampleIF.html
If Statement Example
You are eligible to vote.


127.0.0.1:5500/Practical%2003/MouseUp_Down.html
Mouse Down and Mouse Up Events.
Mouse Button Released



5. Case Study Title

Grading system

6. Case Study Program Code

File 1 : index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Gradesheet</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container">

<h1>GRADESHEET</h1>

<h3>Symbiosis Institute of Technology, Nagpur</h3>
<p><b>Programme:</b> Bachelor of Technology (Computer Science & Engineering)</p>
<p><b>Semester:</b> IV</p>
<p><b>PRN:</b> 24070521001</p>
<p><b>Year:</b>2026</p>

<form id="gradeForm">

<label>Student Name</label>
<input type="text" id="name" placeholder="Enter Name">

<table>

<tr>
<th>Course</th>
<th>Marks (Out of 100)</th>
<th>Grade</th>
</tr>
```

```
<tr>
<td>Theory Of Computation</td>
<td><input type="number" class="marks"></td>
<td class="grade"></td>
</tr>

<tr>
<td>Java</td>
<td><input type="number" class="marks"></td>
<td class="grade"></td>
</tr>

<tr>
<td>Database Management Systems</td>
<td><input type="number" class="marks"></td>
<td class="grade"></td>
</tr>

<tr>
<td>Operating Systems</td>
<td><input type="number" class="marks"></td>
<td class="grade"></td>
</tr>

</table>

<br>

<button type="button" id="calculate">Calculate Result</button>
<button type="reset" id="reset">Reset</button>

</form>

<h2 id="result"></h2>

</div>

<script src="script.js"></script>

</body>
</html>
```

File 2: script.js

```
let marks = document.querySelectorAll(".marks");
let grades = document.querySelectorAll(".grade");

function getGrade(mark){

if(mark>=90)
return "O";

else if(mark>=80)
return "A+";

else if(mark>=70)
return "A";

else if(mark>=60)
return "B+";
```

```
else if(mark>=50)
return "B";

else if(mark>=40)
return "P";

else
return "F";

}

marks.forEach((input,index)=>{

// Keyboard Event
input.addEventListener("keyup",function(){

let value=Number(this.value);

grades[index].innerHTML=getGrade(value);

});

// Prevent invalid keys

input.addEventListener("keydown",function(e){

if(e.key=="-" || e.key=="e"){

e.preventDefault();

}

});

// Mouse Event

input.addEventListener("mouseover",function(){

this.style.backgroundColor="lightyellow";

});

input.addEventListener("mouseout",function(){

this.style.backgroundColor="white";

});

// Validation

input.addEventListener("blur",function(){

if(this.value>100){

alert("Marks cannot exceed 100");

this.value="";
grades[index].innerHTML="";

}

});
```

```
});

let calculate=document.getElementById("calculate");

// Mouse Events

calculate.addEventListener("mouseover",function(){

calculate.style.backgroundColor="blue";

});

calculate.addEventListener("mouseout",function(){

calculate.style.backgroundColor="green";

});

// Click Event

calculate.addEventListener("click",calculateResult);

function calculateResult(){

let total=0;

let count=0;

marks.forEach(function(subject){

if(subject.value!=""){

total+=Number(subject.value);

count++;

}

});

if(count==0){

alert("Enter Marks");

return;

}

let percentage=total/count;

let overall=getGrade(percentage);

document.getElementById("result").innerHTML=

`
Student : ${document.getElementById("name").value}<br>
Total Marks : ${total}<br>
Percentage : ${percentage.toFixed(2)}%<br>
Overall Grade : <b>${overall}</b>

`;
```

```
}

// Reset Button Event

document.getElementById("reset").addEventListener("click",function(){

alert("Form Reset Successfully");

});
```

File 3 : style.css

```
body{

font-family:Arial;
background:#f5f5f5;

}

.container{

width:80%;
margin:auto;
background:white;
padding:20px;
box-shadow:0px 0px 10px gray;

}

h1,h2,h3{

text-align:center;

}

table{

width:100%;
border-collapse:collapse;

}

table,th,td{

border:1px solid black;

}

th{

background:#2b7cff;
color:white;

}

td{

padding:10px;

}

input{

width:90%;
padding:8px;

}

button{

padding:10px 20px;
```

```
margin:10px;
cursor:pointer;
font-size:16px;

}

#calculate{

background:green;
color:white;

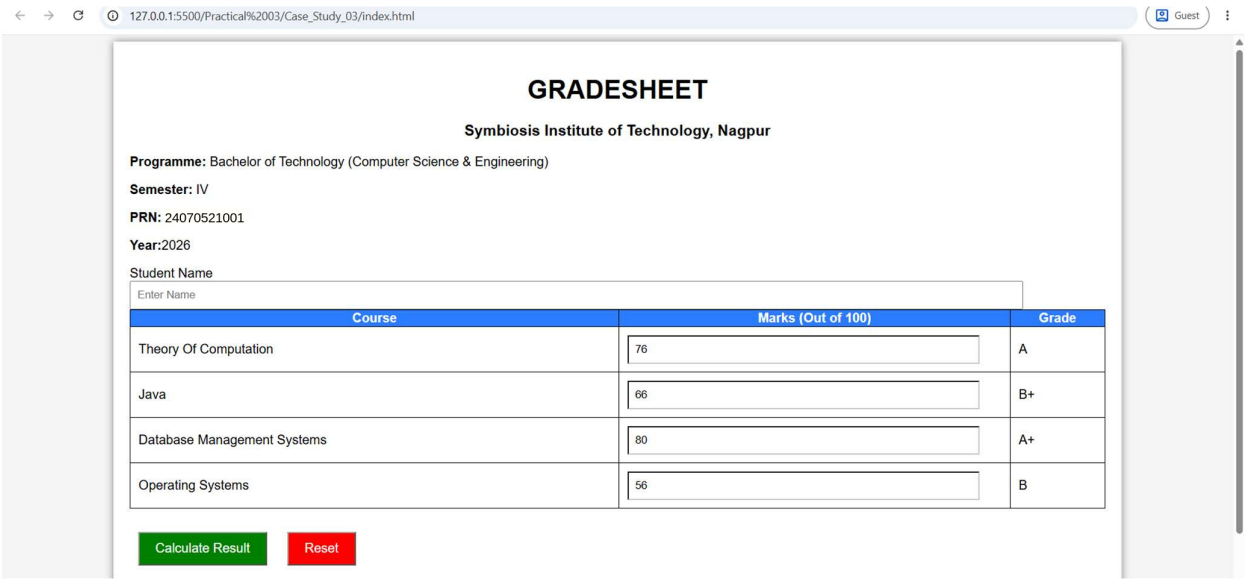
}

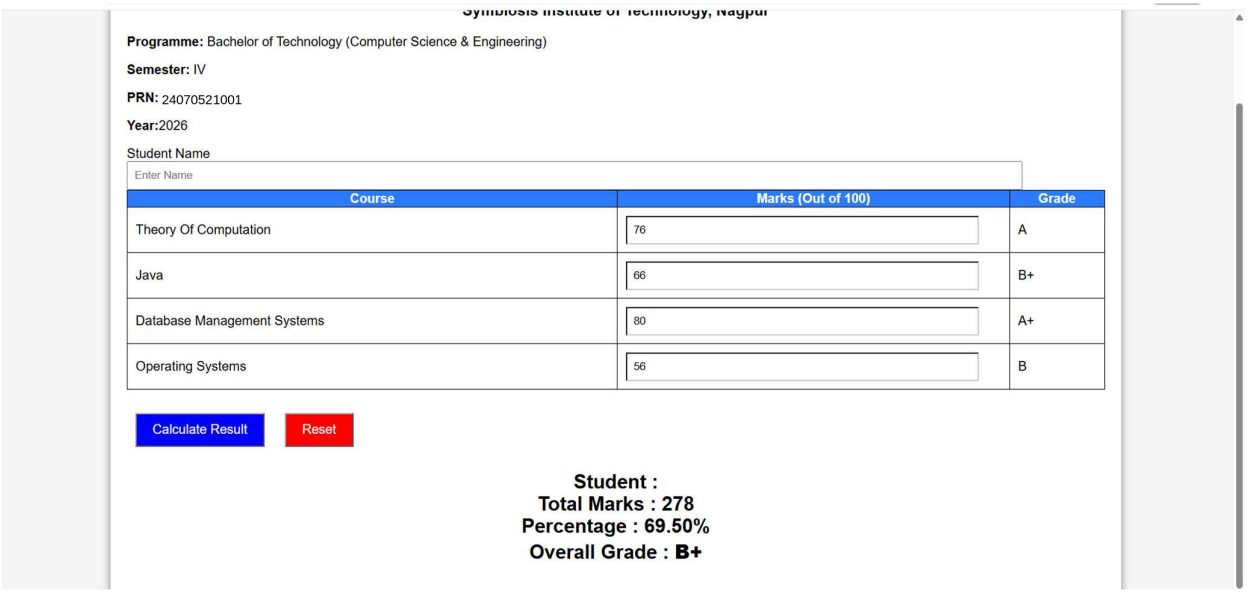
#reset{

background:red;
color:white;

}
```

7. Output of the case study :





8. Result/Conclusion

Thus, control structures (if-else, switch, loops) were successfully used to implement a grading system based on user-entered marks, and addEventListener() was used to handle mouse and keyboard events for a more interactive experience.